Technical Project Summary

Autonomous UAV System for Rapid Disaster Reconnaissance

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Context and Objective

Motivated by recurring wildfire and disaster-response failures in Lebanon (including the 2020 Beirut port explosion), this self-funded, three-person team project set out to design and build a fully autonomous UAV capable of navigating hazardous, GPS-denied, and hard-to-reach environments to detect fire, locate victims, and deliver real-time situational awareness to first responders — without requiring direct human piloting. The project was developed and validated during a period of active regional conflict in Lebanon (2024–2025), which caused frequent GNSS and network disruption and restricted outdoor UAV flight testing, requiring extensive simulation-based validation ahead of constrained real-world flight windows.

Technical Contribution (Technical Lead role)

- As Technical Lead, carried out the majority of the system's design, integration, and implementation, and personally covered the majority of project costs — the entire system was self-funded with no institutional support.
- ROS2-based autonomy architecture integrating flight control, state management, and inter-module communication, bridged to ArduPilot/Pixhawk via MAVROS.
- 360° LiDAR pipeline for real-time obstacle detection and avoidance, validated in RViz simulation and open-field flight tests to ±2 cm tolerance against ground-truth measurements.
- Sensor fusion pipeline combining camera, IMU, and range sensors for robust state estimation and situational awareness (GPS accuracy ±2–3 m in field tests).
- Deep-learning computer vision models trained and deployed on a companion computer (Raspberry Pi) for real-time detection: 92% human detection accuracy (YOLOv8) and 80% fire detection accuracy.
- Full mechanical design (3D CAD) and electronics stack; total assembled weight of 1.4 kg.
- Multi-phase validation: bench-level hardware checks, QGroundControl/Mission Planner mission simulation, and real-world autonomous flight tests (waypoint navigation, obstacle avoidance, return-to-launch), achieving 18 minutes of stable flight endurance.

Validation and Recognition

The system was formally presented to the Lebanese Civil Defence, whose representative attended the FYP defence and confirmed its indispensable value for Lebanon's wildfire response, citing the absence of institutional funding — not the engineering — as the only barrier to real-world deployment. Real-world testing confirmed accurate waypoint navigation, real-time telemetry transmission, and effective autonomous obstacle avoidance, validating the robustness of the integrated ROS2-based architecture for field deployment.

Broader Research Relevance

This project provided hands-on, end-to-end experience across the full autonomy stack a single UAV needs to operate reliably and independently under real-world uncertainty — including active GNSS-denied conditions — combining LiDAR-based obstacle avoidance, sensor fusion, learning-based perception, and autonomous navigation. This forms a direct technical foundation for extending single-agent autonomy to coordinated multi-robot systems, robust control under uncertainty, and autonomous decision-making architectures for search-and-rescue, reconnaissance, and other real-world robotics applications.

Project Repository

Full documentation, system architecture, and results: github.com/wadih553/autonomous-uav-disaster-recon